Toronto Home-Price Prediction

Off-market

What is this home worth before listing?

No asking price

List-aware

What will this listing close at?

Asking price available

150,000 synthetic Toronto sales • Jan 2021–May 2026 • No outside prediction data

Official monthly and team fixed-four-month validation

Purpose	Training cutoff	Evaluation / forecast	Homes
Official monthly	Before each month	Dec 2025–May 2026	10,069
Team fixed four-month	Before Feb 1, 2026	Feb–May 2026; no updates	7,773
Final refit	Before Jun 1, 2026	Jun–Sep 2026; hidden labels	3,974 / 4,064

FEATURE LOOKBACK

90 / 180 / 365 days

Past-market summaries

TRAINING SAMPLE

Several years

Expanding eligible past sales

FORECAST HORIZON

One or four months

Monthly refits or one frozen batch

No current-fold outcomes enter fitting or history pools.

Historical queries end before min(listing date, cutoff). Settings stay fixed. Configuration selection overlapped Dec 2025, so these remain cumulative development results.

LightGBM: efficient gradient-boosted trees

A sequence of decision trees for tabular data.

1 Boosting: learn corrections

Each new tree learns a loss-based correction to the score.

2 Leaf-wise: split where it helps most

Grow the leaf with the largest expected loss reduction.

3 Histograms: search splits efficiently

Group numeric values into bins to reduce search cost.

Why it fits our task

Nonlinear relationships

Location, size and type can interact.

Numeric and categorical inputs

Property attributes and local market context.

Controlled tree complexity

Leaf limits, minimum samples and penalties.

$$\underbrace{f_m(X)}_{\text{updated score}} = \underbrace{f_{m-1}(X)}_{\text{current score}} + \eta \underbrace{g_m(X)}_{\text{new tree correction}}$$

η: learning rate. Prediction combines the frozen trees; no per-home retraining.

Leaf-wise growth can overfit; complexity constraints matter. Mechanism: LightGBM official feature guide. Task suitability is our modelling rationale, not proof of superiority.

From tree scores to home prices

TRAINING POPULATIONS

Off-market

Related types share examples; separate routes allow different price relationships.

List-aware

One global model shares information across types and allows type interactions.

Off-market

Property-type routes: LightGBM + CatBoost

$$t = \log P \qquad \widehat{P} = \exp(f(X))$$

List-aware

One global LightGBM across all six types

$$t = \log(P/L) \qquad \widehat{P} = L \exp(f(X))$$

P: sale price L: asking price X: available features

Shared loss: absolute error in log space

$$\mathcal{L} = \frac{1}{N} \sum_i \left| f(X_i) - t_i \right|$$

Relative-error surrogate; extreme residuals have less influence than under squared loss.

Related types share examples. Outputs rounded. L1 is a surrogate; no loss ablation establishes optimality.

Results under both validation protocols

Official monthly: Dec 2025–May 2026. Team fixed four-month: Feb–May 2026.

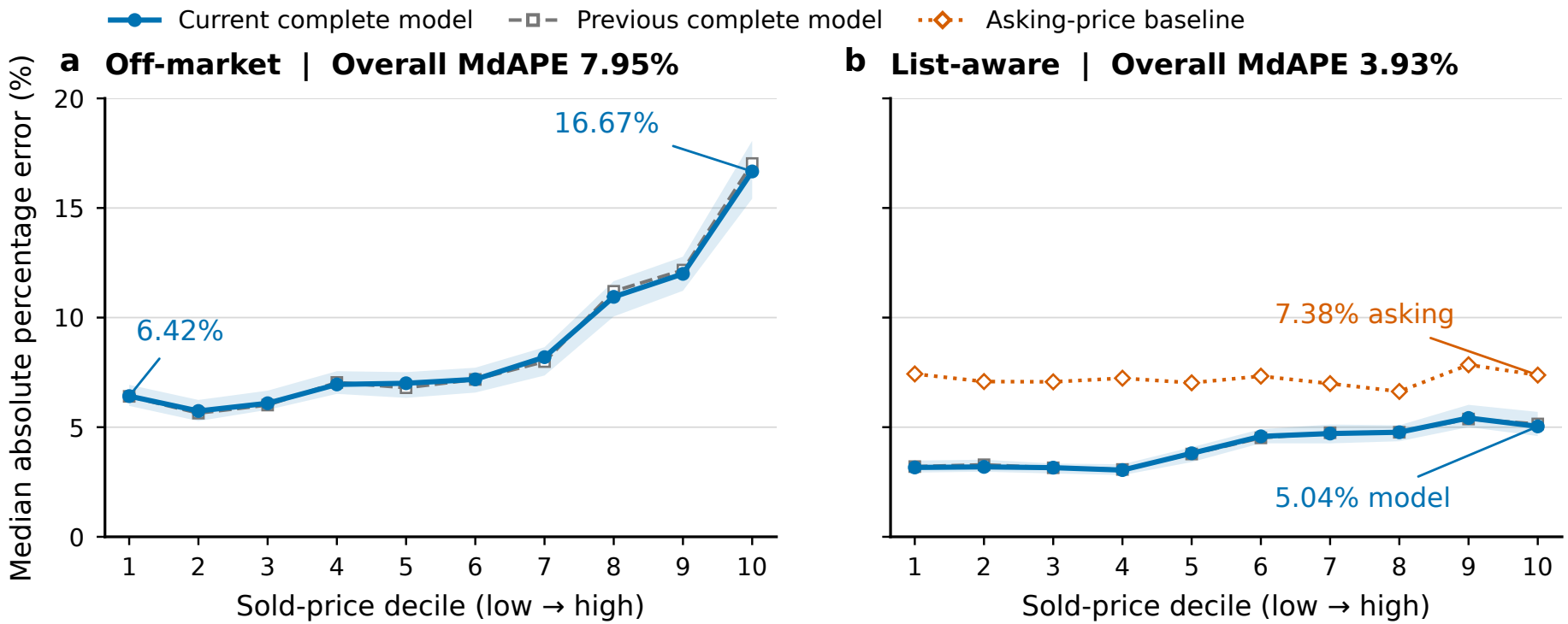
Track / model (validation protocol)	Homes	MdAPE	MAPE
Off-market trees (official monthly)	10,069	**7.62%**	10.19%
List-aware asking price (official monthly)	10,069	7.26%	8.09%
List-aware ratio (official monthly)	10,069	**3.93%**	5.04%
Off-market trees (team fixed four-month)	7,773	**7.95%**	10.38%
List-aware asking price (team fixed four-month)	7,773	7.20%	8.03%
List-aware ratio (team fixed four-month)	7,773	**3.93%**	5.19%

Asking-price baseline = use the listed price as the prediction.

Same selected specifications. Cumulative development validation.

MdAPE is median individual percentage error; MAPE is the mean. Lower is better. Compare baselines within each protocol. Different samples, cutoffs and updating prevent a direct ranking across protocols. Hidden-test error is unknown.

High-price homes remain the hardest to value



Frozen February–May 2026 development backtest; N = 7,773; n = 775–780 per decile.

Shading: pointwise 95% bootstrap CI. Realized-price deciles are retrospective diagnostics only.

Use extra review for expensive or unusual homes.

Use the estimate with its limits

Before listing	Off-market supports initial valuation.
After listing	List-aware adjusts the asking-price anchor.
Extra review	Expensive, unusual or weakly supported homes.

Next: unused-period testing and past-only uncertainty intervals.

Official monthly MdAPE: 7.62% / 3.93%

Hidden-test accuracy unknown. Fitted explanations describe model behaviour, not causal effects.